# COMPUTATIONAL MODELLING OF SALMON MIGRATION IN THE TORNIONJOKI RIVER

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## ABSTRACT

The aim of this project was to develop a computational model to estimate annual Atlantic salmon (*Salmo salar*) migration in the Tornionjoki River. Salmon fishing tourism in the Tornionjoki River is estimated to be worth around ten million euros. Salmon are also ecologically important to the Baltic Sea. A simplified state-based model was constructed from the salmon life cycle with fishing and predation by seals (*Halichoerus grypus*). A computational model based on the Leslie population model was implemented in the Rust programming language using a state-based model. The model was fitted to observed data on Tornionjoki salmon migration. The resulting model reproduces migration patterns from 2017 to 2022 and captures the overall fluctuations in salmon abundance in the Tornionjoki River.

## 1. THE PROBLEM

The Tornionjoki river is an important river for recreational and professional fishing. The fishing tourism of Tornionjoki is worth 10.8 million euros according to a study done by the University of Helsinki.[3] The fishing culture of Tornionjoki river is also important to the locals of Lapland.

Salmon *(Salmo salar)* migration has significantly decreased in the years 2023 and 2024 as seen in Figure 1. Salmon migration increased in the summer of 2025, but most of the salmon were male, which don't produce more young salmon.[2] The reason for the sharp decline of migrating salmon in recent years is not yet known.[6]

The study examines salmon migration by creating a **computational model** to represent the fluctuations of migrating salmon for the years 2009–2025. The model was used to predict future salmon migration and estimate the impact of fishing and gray seals (*Halichoerus grypus*) on salmon migration.

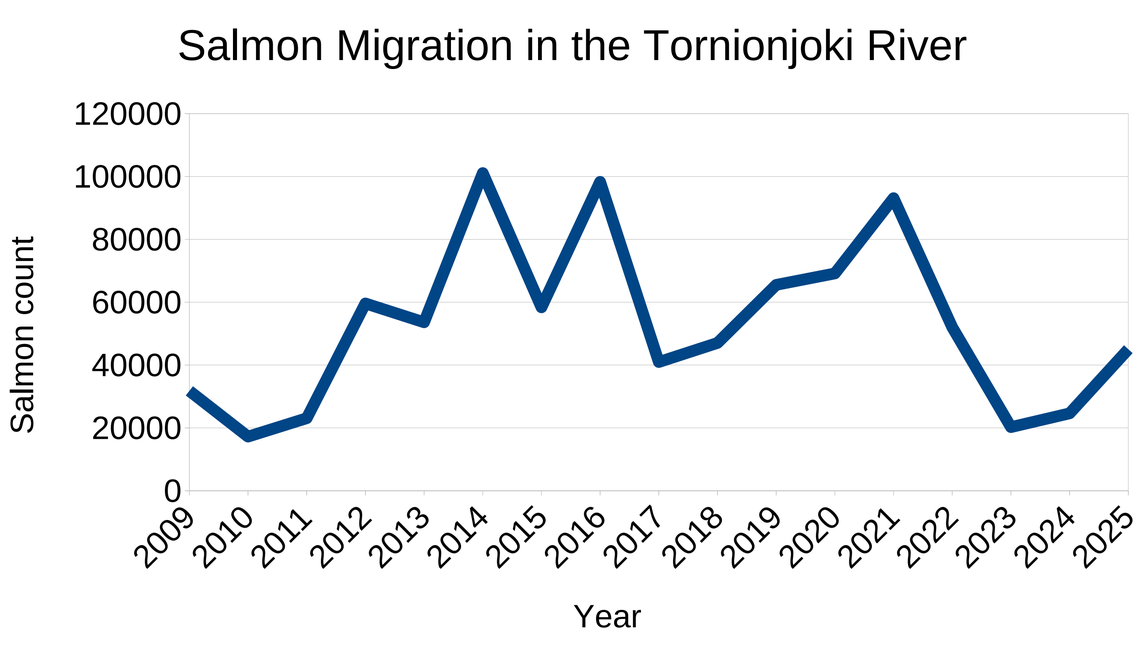



***Figure 1:*** *Yearly salmon migration in the Tornionjoki river.*

## 2. SALMON LIFECYCLE

**1. EGG**
The Atlantic salmon's life cycle begins in freshwater rivers.

**2. ALEVIN**
Alevin stay hidden in the gravel nest and feed on their nutrient-rich yolk sac.

**3. FRY**
After hatching from the yolk sac, fry hunt different insects in the river.

**4. PARR**
Mortality decreases after the first summer of the juvenile salmon. During this phase they eat bigger prey.

**5. SMOLT**
The phase right before the sea. Behavioural and physiological changes occur here.

**6. ADULT**
Adults feed in the sea before returning to their natal river to spawn.

In the study the Atlantic salmon life cycle is modelled after Tuomas Manelius's master's thesis.[1] Atlantic salmon are an anadromous species whose life cycle begins in freshwater rivers, after which juveniles migrate to the sea, where they feed before returning to their natal river to spawn. Figure 2 shows arrows depicting the salmon life cycle in the Tornionjoki river bordering Finland and Sweden.

Only a fraction of salmon survive to reproduce. Fishing is present in the sea and the river and predation by seals occurs in the sea and near the mouth of the Tornionjoki river.

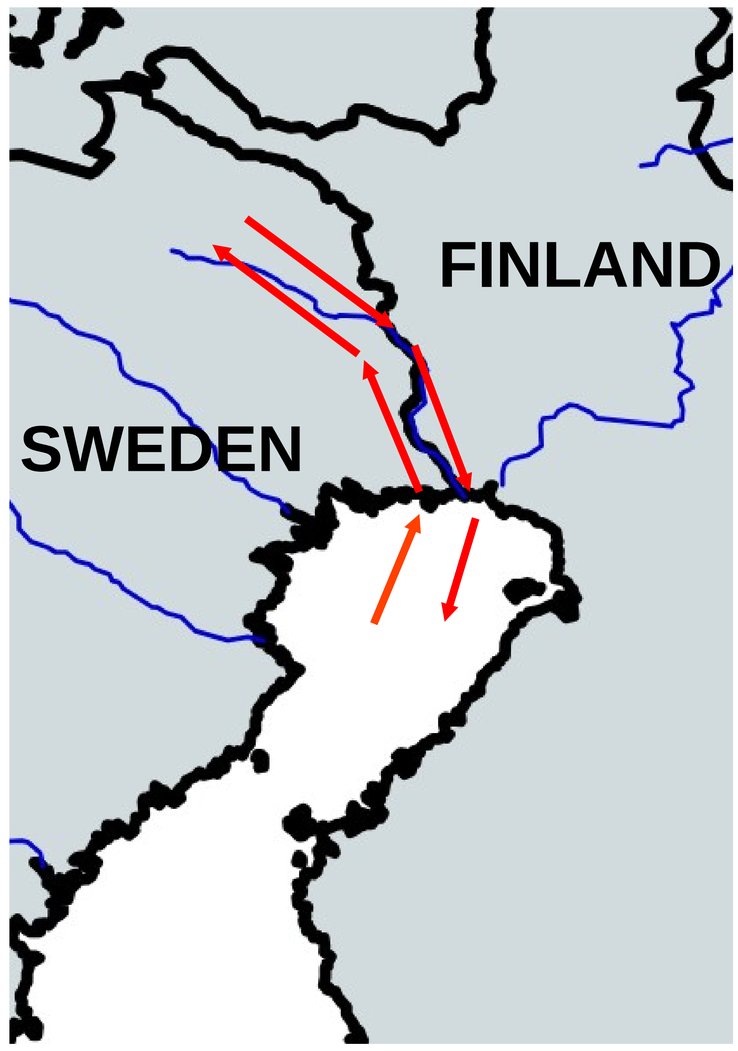



***Figure 2:*** *Image of the Tornionjoki river with arrows depicting the salmon life cycle.[4]*

## 3. SIMPLIFIED MODEL

A **simplified state-based model** was constructed from the salmon life cycle with fishing mortality and predation by seals.

Figure 3 shows that the **simplified model** is split into three different phases: freshwater, sea-phase salmon and spawning salmon. The freshwater phase lasts 2–5 years in the Tornionjoki river and each sea-phase and spawning phase lasts for one year.

The mortality factors were assumed to be yearly gray seal populations and fishing. The mortality rates of freshwater salmon and spawning salmon were represented as transition probabilities in the **state-based model**. The transition probabilities of the **state-based model** were collected from previous studies. Transition probabilities that weren't found from previous studies were estimated computationally.

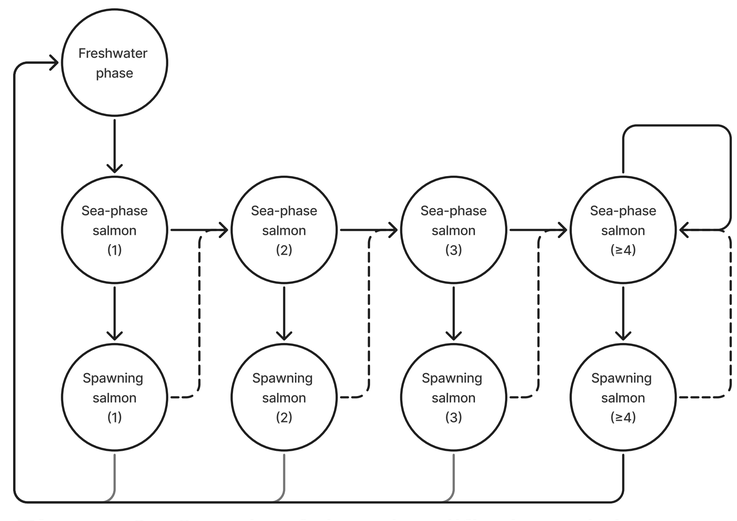



***Figure 3:*** *Graph of the simplified model.*

# 4. COMPUTATIONAL MODEL

A **computational model** was written in the Rust programming language to describe salmon population fluctuations and survival. The computational model is based on the **Leslie population model** and the **state-based model** of the salmon life cycle.[5]

Let $n_t$ be the salmon population vector at time and $L$ be the Leslie matrix. Thus the **modified Leslie model** is $n_{t+1}=L(n_t-\alpha\cdot h_t-k_t)$ which is expanded in Equation 1.

The first row of the Leslie matrix has the fecundities of the population. The other elements of the matrix contain the transition probabilities from one age class to another.[5]

Before the state transition, the population vector is decreased by the yearly fishing amount, which includes sea fishing and river fishing. It is also decreased by the mortality produced by gray seals which was assumed to be proportional to the seal population. Mortality caused by seals and fishing is distributed proportionally to the number of males and females.

$$\begin{bmatrix} n_0 \\ n_1 \\ \vdots \\ n_{\omega-1} \end{bmatrix}_{t+1} = \begin{bmatrix} f_0 & f_1 & \dots & f_{\omega-2} & f_{\omega-1} \\ s_0 & 0 & \dots & 0 & 0 \\ 0 & s_1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & s_{\omega-2} & 0 \end{bmatrix} \left( \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_{\omega-1} \end{bmatrix}_t - \alpha \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_{\omega-1} \end{bmatrix}_t - \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_{\omega-1} \end{bmatrix}_t \right)$$

***Equation 1:*** *The modified Leslie model based on the simplified state-based model.*

## MODEL FITTING

The parameters of the **population model** were fitted to the observation data of the Tornionjoki river's salmon runs from years 2009 to 2025. The parameters' values were estimated by maximizing the **coefficient of determination** between the calculated salmon runs and the measured salmon run data. The predictions of the parameter combinations that gave the best **coefficient of determinations** were checked manually. The parameter combinations that matched realistic values were chosen for the final **computational model**.

## MODEL PREDICTION

The **computational model** was used to predict salmon migration up to the year 2030. Variation of either the number of gray seals or fishing was done in Figures 6 and 7 while the other number was constant.

The seal population's growth was assumed to grow linearly and fishing to stay constant in the main prediction.

# 5. RESULTS AND CONCLUSIONS

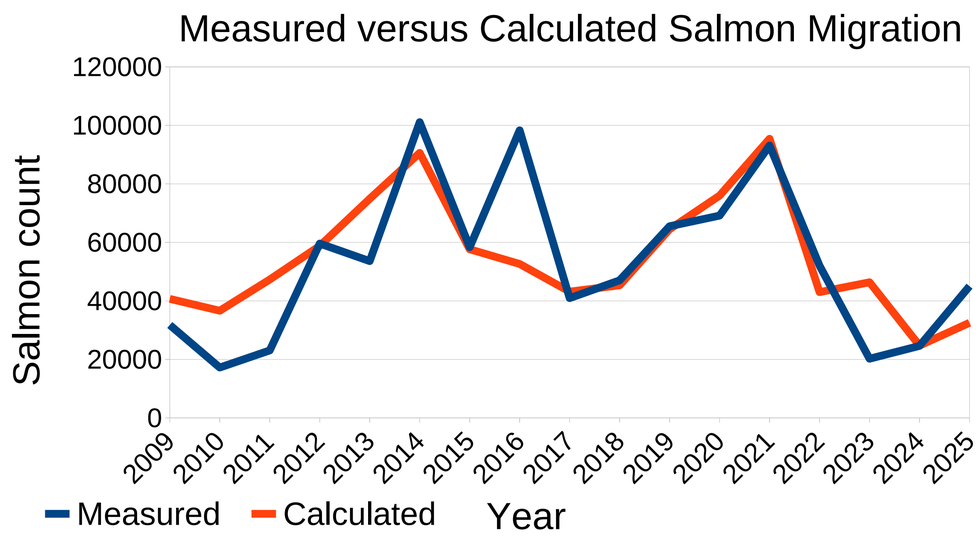



***Figure 4:*** *The calculated salmon runs compared to the measured salmon runs in the Tornionjoki river for the years 2009–2025.*

## 1. BEST FITTED MODEL

The **computational model** was able to explain the population fluctuations of 2017–2022 well, but it had significant error with earlier years as seen in Figure 4. According to the model, the weak salmon migration in recent years is because of natural population fluctuations. The low production of young salmon of the previous weak years can be seen with a delay in 2023 and 2024.

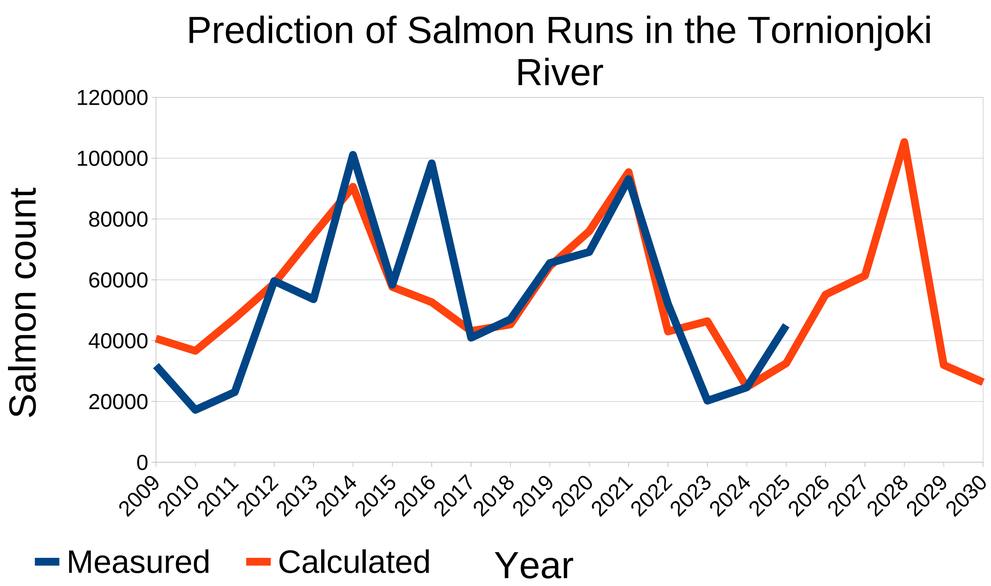



***Figure 5:*** *The main prediction of the Tornionjoki river's salmon runs.*

## 2. PREDICTION

The model predicts a similar curve compared to the measured data for the years 2023–2025 with a one year delay in Figure 5. This might be because of 2025's exceptional amount of young male salmon.[2] According to the prediction while taking into account the one year delay, the Tornionjoki river's salmon migration will increase in 2026 and peak at 2027.

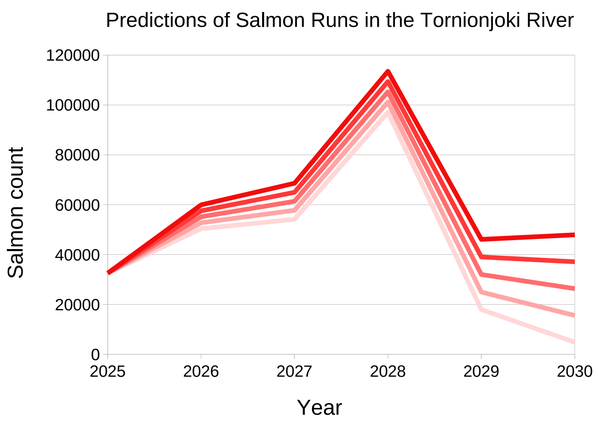



***Figure 6:*** *Variation in the number of gray seals.*

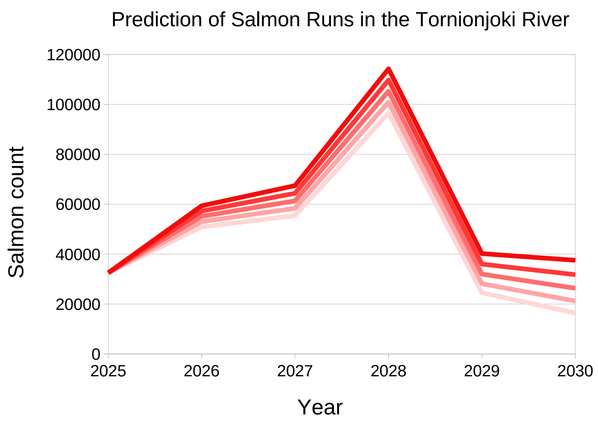



***Figure 7:*** *Variation in fishing.*

## 3. SEALS AND FISHING

People can affect salmon migration by reducing fishing and hunting gray seals. The effects of these have been estimated by the predictions in Figures 6 and 7. The lightest curve represents the most amount of seals or fishing and the darkest curve represents the least amount of seals or fishing in Figures 6 and 7. According to the model the effect of fishing and gray seal population changes can be seen with a delay of five years.

# MODEL ACCURACY

A sensitivity analysis was made for the **computational model** by calculating results while the inputs were varied. The variation was done using a nearly orthogonal Latin hypercube. The calculated results for each parameter were gathered into scatter plots.

The result was that the model is very sensitive to its parameters. Change in fecundity either leads to exponential growth or decline in salmon migration. According to the **computational model** the removal of both fishing and seals leads to strong growth to biologically impossible levels. However the salmon population's growth is limited by other biological factors such as the quantity of food when seals and fishing are removed.

In addition to the results of the sensitivity analysis the computational model lacks historical data and its accuracy is based on only 15 years of measured salmon migration data. The model also has many presumptions such as a constant fecundity that decrease the accuracy of the computational model.

The model can follow the Tornionjoki river's salmon population fluctuations but it has more error in the beginning. This may be due to a lack of historical data, which reduces the model's predictive capability during the initial years of the measured data (Figure 8).

Biological constraints such as food availability should be added to the model. More historical data should be accumulated to be able to create a better model and gender related differences should be considered. The model could also be integrated into a larger framework to study relationships such as those between gray seals and Atlantic salmon.

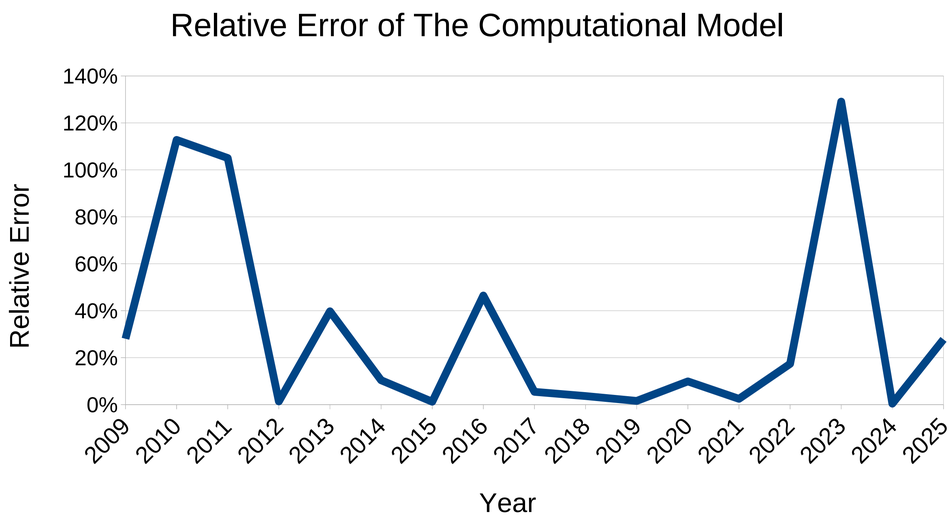



***Figure 8:*** *Relative error between the calculated salmon migration and the measured salmon migration.*

# MORE ON THE COMPUTATIONAL MODEL

***Table 1:*** *Estimated parameters of the computational model.*

| Parameter | Description | Estimated value |
|---|---|---|
| $\mathbf{P}_L$ | The probability that freshwater phase salmon migrate to the ocean each year. Young salmon migrate to the ocean after 2–5 years. | 0.1 |
| $\mathbf{P}_J$ | The probability that a salmon stays in the ocean and doesn't ascend the river every year. | 0.2 |
| Alpha | Scales the mortality produced by seals in the Leslie model. | 0.6 |
| Gamma | The amount of salmon from other stocks in the Baltic sea. Scales sea fishing. | 420000 |
| Fecundity | How many eggs a female salmon lays. | 600 |

$$\text{Sea fishing weight} = \frac{L_s}{L_s + \gamma}$$

***Equation 2:*** *Sea fishing is multiplied by the proportion of Tornionjoki salmon in the sea.*

## STARTING POPULATION VECTOR

The age distribution of the model's salmon population in the beginning is estimated by calculating the stable age structure of the Leslie matrix. This is done by simulating the Leslie model for 100 years without fishing and seal induced deaths. The resulting population vector is normalized and scaled to correspond with first year salmon run data. Then the program runs the Leslie model once with year one data to simulate seal induced unbalance in the age distribution.

## PARAMETERS

Transition probabilities $\mathbf{P}_J$ and $\mathbf{P}_L$ were chosen as parameters, because similar values were not found from previous studies. The mortality caused by gray seals was estimated using the alpha parameter. Fecundity was chosen as a constant parameter to simplify the model. The parameters of the model are in Table 1.

The gamma parameter was chosen to calculate sea fishing because the Baltic sea salmon fishing material includes salmon from other river stocks and hatchery-reared salmon. Before sea fishing is applied the program multiplies it by the factor in Equation 2, where $L_s$ is the amount of Tornionjoki salmon in the sea and $\gamma$ (Gamma) is the amount of salmon from other stocks in the Baltic sea.

## SEALS AND FISHING IN THE MODEL

Mortality caused by seals and fishing is distributed proportionally to the number of males and females. Because there are more gray seals in the Bothnian Bay compared to the Baltic Sea's main basin, seal induced mortality is split between sea-phase salmon and spawning salmon. Eighty percent of the mortality caused by seals is split to spawning salmon and twenty percent to sea-phase salmon because there are more seals near the Tornionjoki river's mouth and they eat more salmon.

The first year data of fishing was left out of the input because it differed from later years. Analysis showed that the first year fishing data caused instability to the model. Therefore due to the analysis and the lack of historical data, the input was narrowed to later, more comparable years.

# IMPORTANT REFERENCES


[1] T. Manelius. TORNIONJOEN LOHEN NOUSUN AJOITTUMINEN SUHTEESSA LOHIEN KOKOON JA KALASTUKSEEN. Pro gradu -tutkielma. Itä-Suomen yliopisto. Ympäristö- ja biotieteiden laitos. 2020

[2] Yle Uutiset. Tornionjoen lohikesä näytti hiljaiselta, mutta sitten tapahtui jotain erikoista: paikalle tupsahti runsaasti koiraita. https://yle.fi/a/74-20176542

[3] "Erätalouden kehittäminen Suomessa -tutkimus | Ruralia-instituutti | Helsingin yliopisto." Accessed: Jun. 25, 2026. [Online]. Available: https://www.helsinki.fi/fi/uutiset/matematiikka-ja-luonnontieteet/eratalouden-kehittaminen-suomessa-tutkimus

[4] "World Map - Simple | Create a custom map," MapChart. Accessed: Sep. 11, 2026. [Online]. Available: https://mapchart.net/world.html

[5] P. H. Leslie, "On the Use of Matrices in Certain Population Mathematics," Biometrika, vol. 33, no. 3, pp. 183–212, 1945, doi: 10.2307/2332297.

[6] "How is the situation for Baltic salmon? | slu.se." Accessed: Jun. 25, 2026. [Online]. Available: https://www.slu.se/en/news/2025/01/hur-mar-laxen-i-ostersjon/